

INDEX

Analog Circuit

[illegible]

LECTURE NO : 03

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Date _____

Page 9

Current Density (J_n)

$$J_n = \mu_n E n q \quad \text{A/cm}^2$$

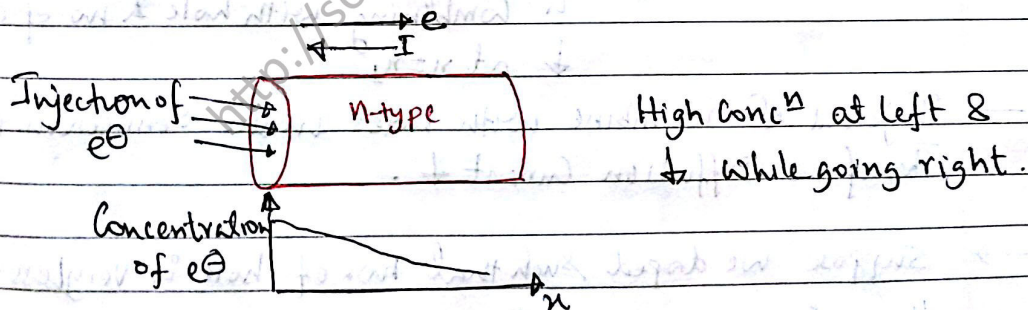
$$\begin{aligned} \text{Total Current Density} &= J_n + J_p \\ &= (\mu_n n q + \mu_p p q) E \\ &\approx \begin{cases} 400 \text{ cm}^2/\text{volt-sec} \\ 1350 \text{ cm}^2/\text{volt-sec} \end{cases} \end{aligned}$$

Lecture : 03

2nd Mechanism that does not require a Voltage or an electric Field called Diffusion.

we could have Current without voltage (violates ohm's law).

→ Movement of charge carriers from region of high concentration to Low concentration is known as Diffusion.



→ If there is no injection, there is no diffusion. In other way if slope = 0 then there is no diffusion.

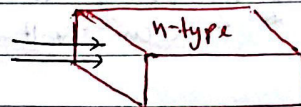
$$\therefore J \propto \frac{dn}{dx}$$

$$J = \underbrace{D_n}_{\text{Diffusivity}} \frac{dn}{dx} q$$

In General

$$J_{\text{total}} = \left(\underbrace{D_n \frac{dn}{dx}}_{\text{due to } e^-} + \underbrace{D_p \frac{dp}{dx}}_{\text{due to holes}} \right) q$$

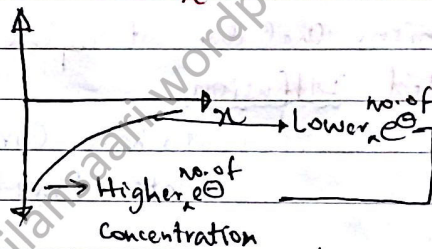
$$D_n = 34 \text{ cm}^2/\text{sec} \quad D_p = 12 \text{ cm}^2/\text{sec}$$

ExampleInjection of e^- 

Plot the resulting diffusion current



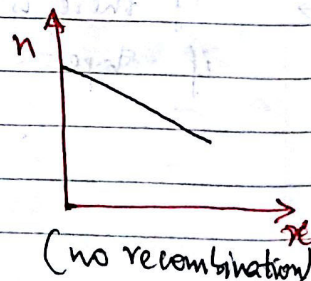
J

Higher no. of e^- concentrationLower no. of e^- Where e^- are going?Conclusion

As e^- is moving from $x=0$ to $x=x_1$, it is combining with hole & no. of e^- (free) ↓ at $x=x_1$.

→ Injected e^- recombine with holes in the Semiconductor. Therefore Diffusion Current ↓.

→ Suppose we doped such that no. of hole is very less i.e. there is no recombination. Hence graph will look like



(no recombination)

Einstein Relation :

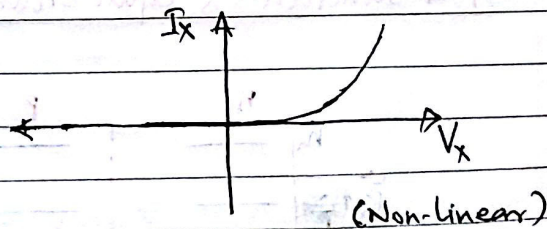
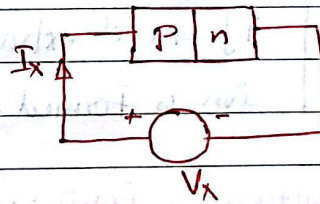
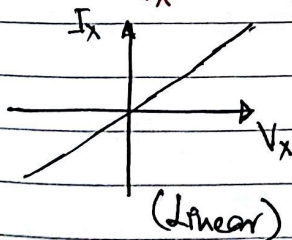
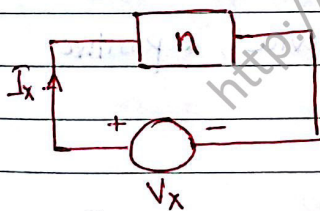
$$\frac{D}{\mu} = \frac{KT}{q}$$

→ 26mV at $T=300K$.
→ Thermal voltage V_T

Now we are ready to build the first electronic devices known as P-N Diode.

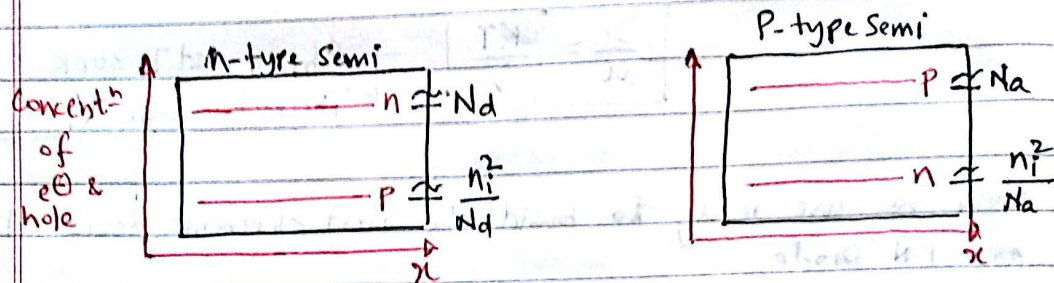
PN JUNCTIONApplication

- chargers & Adapters
- Voltage multipliers
- Rectifier.

Quick ExperimentTwo questions:

- ① How do the charge carriers redistribute themselves after formation of PN junction?
- ② How does PN junction behave under ~~for~~
 - equilibrium condition
 - Reverse biased & forward bias condition.

Some Observation



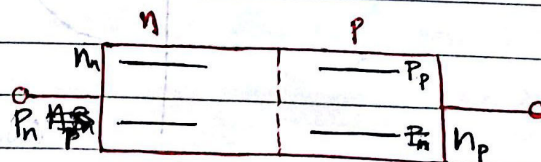
Charge Neutrality

n → What is the Net charge in this?

→ Every e^- present here have one counterpart proton in the Nucleus i.e. if e^- available from phosphorus atom, that phosphorus atom have proton inside the Nucleus. If e^- available from Si atom, that Si atom also have proton inside the Nucleus. So, the total charge in this semiconductor is Zero.

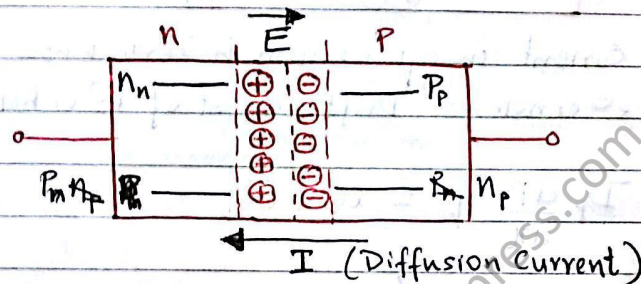
$n \oplus$ If one e^- extracted from device, a positive ion is Formed.

* P-N Junction in Equilibrium



→ We have high concentration of e^- in n & Low concentration of e^- in p. Therefore Diffusion takes place. Similarly For p we have high concentration of hole in p & less concentration of hole in n. Therefore Diffusion takes place.

- In diffusion process, e^- from n goes to p & holes from p goes to n type Semiconductor.
- Before attaching p & n its neutral, but when one e^- moves from n to p; it left one +ve ion. Similarly one hole moves from p to n due to diffusion it left one -ve ion behind.



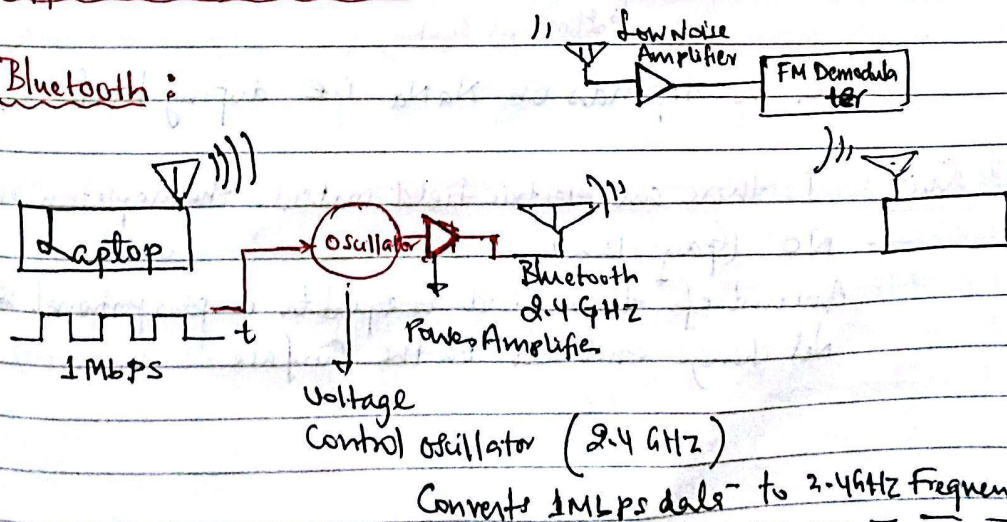
Without giving 'V' I Formed due to diffusion.

- Because of +ve & -ve ions at the junction electric field produced. Now, if e^- from n want to go to p then -ve ion from p oppose them means electric field wants to stop them. Finally diffusion stops. At some point diffusion current = 0 & PN Junction comes to an equilibrium. The region formed by +ve & -ve ion is known as depletion region.

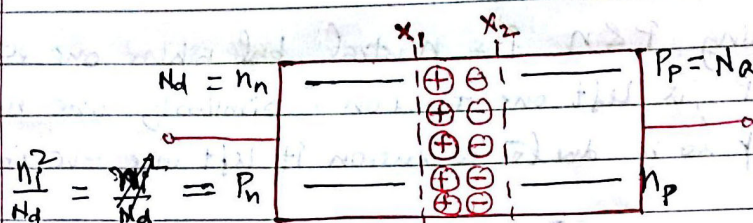
Lecture: 04

Application of Electronics:

Bluetooth:



PN Junction in Equilibrium ... Continued



→ The overall current in equilibrium is Zero when diffusion current of e^- or hole = Drift current of e^- or holes.

$$D_p \frac{dp}{dx} q = \mu_p p E q$$

$$\int_{p_n}^{p_p} D_p \frac{dp}{p} = \int_{x_1}^{x_2} \mu_p E dx$$

$$D_p \ln \frac{p_p}{p_n} = \mu_p [V_{x_1} - V_{x_2}]$$

$$V_{AB} = - \int_A^B E \cdot dx$$

$$D_p \ln \frac{N_a}{n_i^2/N_d} = \mu_p V_0$$

$$V_0 = \frac{D_p}{\mu_p} \ln \frac{N_a}{n_i^2/N_d} = \frac{kT}{q} \ln \frac{N_a N_d}{n_i^2}$$

Built-in

Potential

$$V_0 = \frac{kT}{q} \ln \frac{N_a N_d}{n_i^2}$$

↳ 0.6V at 300K

So, V_0 depends on $N_a N_d$ i.e. doping level.

Quiz: Is there an electric field outside the depletion region?

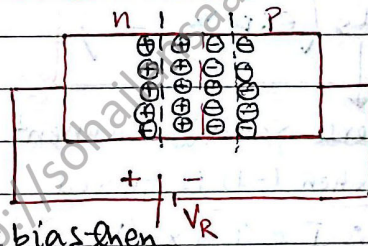
→ No (Gauss law)

Amount of electric field is equal to is proportional to Net charge enclosed on the surface

Observation:

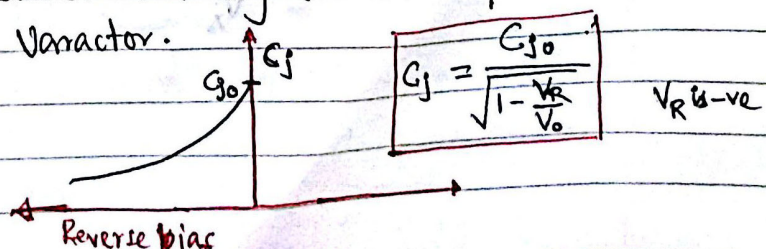
- Same V_0 expression is obtained if the e^- currents are considered.
- If N_A & N_D are in the order of $10^{16}/\text{cm}^3$ then

$$V_0 = 26\text{mV} \ln \frac{10^{32}}{10^{20}} = 720\text{mV} = 0.72\text{V}$$
- Typical values of V_0 are from (700-800) mV.
- V_0 is localized (within depletion region)
- We cannot measure V_0 from outside. Means if we take Multimeter & connect it to terminal of diode then we get $V = 0\text{V}$ why? because $V_0 = 0.7\text{V}$ does not reach to the terminal of diode due to which we cannot measure like this.

PN Junction in Reverse bias

→ When we Reverse bias then +ve terminal of battery attracts some e^- from n region means more +ve ions left behind. Similarly -ve terminal attracts some hole from p region which means more -ve ions left behind. Hence depletion region increases.

→ When we move from equilibrium to reverse bias then the capacitance of the junction ↓ due to ↑ in depletion width. Therefore it is called Electronically Variable Capacitor & this device is known as Varactor.

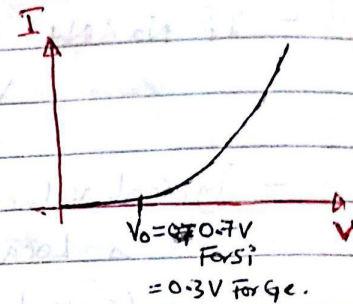
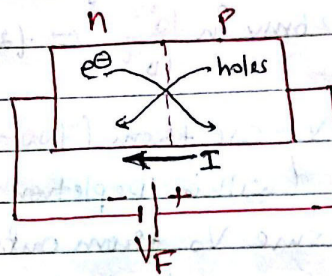


LECTURE : 05

Date
Page

Lecture No: 05

PN Junction in Forward Bias



→ +ve terminal of V_F pushed the holes of P and -ve terminal of V_F pushed the e^- of n region. Both e^- & hole are responsible for Forward Current. Expression of Current in forward bias region is given by

$$I = I_s \left(\exp \frac{V_x}{V_T} - 1 \right)$$

Reverse Saturation Current

$V_x = \text{Input voltage}$

$V_T = V_0 = \text{Terminal voltage}$

$$= \frac{kq}{T}$$

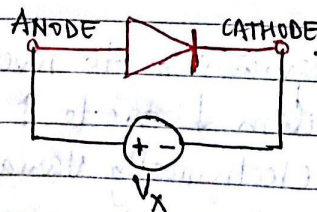
If $V_x \gg V_T$ then (-1) is ignored

~~→~~

$$\therefore I = I_s \exp \left(\frac{V_x}{V_T} \right)$$

DIODE

→ Two terminal device that carries a current when its voltage is +ve & no current when its voltage is -ve. A PN Junction is an example of diode.

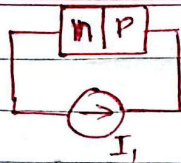


If $V_x > 0$ F.B.

$V_x < 0$ R.B.

Quiz

$$-V_x +$$



$$V = ?$$

$$I_1 = I_s e^{V_x/V_T}$$

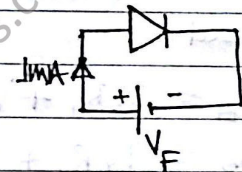
$$\Rightarrow V_x = V_T \ln \frac{I_1}{I_s} \quad \checkmark$$

Example (a) If $I_s = 10^{-16} \text{ A}$ & we measure a forward bias current of 1 mA . What voltage applied to diode?

$$V_F = V_T \ln \frac{I}{I_s} \quad \begin{matrix} \leftarrow 1 \text{ mA} \\ \leftarrow 10^{-16} \text{ mA} \end{matrix}$$

$$V_F = 778 \text{ mV}$$

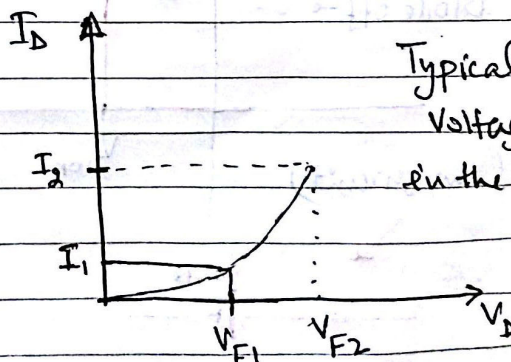
$\hookrightarrow$ This much voltage is applied to the diode.



(b) How much should be voltage increase to raise the current by a factor of 10?

$$\left. \begin{aligned} V_{F1} &= V_T \ln \frac{I_1}{I_s} \\ V_{F2} &= V_T \ln \frac{I_2}{I_s} \end{aligned} \right\} \begin{aligned} V_{F2} - V_{F1} &= V_T \ln \frac{I_2}{I_1} \\ V_{F2} - V_{F1} &= V_T \ln 10 \end{aligned}$$

*** For $10\times$ $\uparrow$ in current we need to increase voltage by $V_T \ln 10 \approx 60 \text{ mV}$.



Typical forward biased voltages for diode are in the range of $700\text{--}800 \text{ mV}$.